

Baseline Comparison Targets for a Bangladesh-Focused Mobile Security Paper (Q1 Security + HCI Venues, 2023–2025)

For a Bangladesh-focused mobile security fusion system, the publishable 2023–2025 comparison set is now well-defined and stable: benchmark detection against **URLTran**, **URLBERT/urlBERT**, **DomURLs_BERT** and **PhishGNN** (URL/phishing); **BanglaBERT**, **mBERT** and **XLM-RoBERTa** plus the hybrid **BERT+CharCNN** and **BERT+XGBoost Bangla** smishing models (low-resource SMS); and **Drebin**, **MaMaDroid**, **DroidEvolver** and **Chen-Ding-Wagner's continuous-learning detector** (Android malware) — all evaluated under the **TESSERACT** temporal/spatial protocol. The single most directly comparable prior system, and the one to beat, is **SmishX (SOUPS 2025)**, which fuses LLM-based smishing detection with human-centered explanation.

TL;DR

- **The detection baselines are settled.** URL: URLTran, URLBERT, DomURLs_BERT, PhishGNN, KnowPhish. Bangla/low-resource SMS: BanglaBERT, XLM-RoBERTa, mBERT, BERT+CharCNN. Android: Drebin, MaMaDroid, DroidEvolver, Chen et al. (USENIX Sec 2023). Evaluation regime for zero-day/drift claims: TESSERACT (temporal + spatial constraints, report AUT), increasingly cross-checked against LAMDA/Transcendent benchmarks.
- **SmishX (Wang et al., SOUPS 2025) is your central comparison** on both axes — it fuses an LLM detector with explanations (98.8% accuracy vs. PhishE 85.0% and SpaLLM-Guard 96.8%) and a usable-security study (N=175, SUS 82.6, detection accuracy 0.712→0.928). It is English/U.S.-only; the **Bangla, low-literacy, externally-grounded, explanation-producing** variant is your defensible novelty gap, alongside the Korean-only KorSmishing explainer (EMNLP 2024).
- **Reviewers treat "genuinely novel" as a new mechanism, a new realistic evaluation regime, or a genuinely under-served population/language — not stacked off-the-shelf parts.** A multimodal fusion that merely concatenates known models is an "engineering combination"; novelty requires a new fusion mechanism, robustness under obfuscation/adversarial/temporal drift, or a new language/population validated with a real user study.

Key Findings

1. URL / phishing detection baselines (the published external systems Q1 papers compare against)

- **URLTran** — Maneriker, Stokes, Lazo et al., *MILCOM 2021* (arXiv 2106.05256). Transformer (BERT/RoBERTa) over raw URLs. The headline benchmark, reported verbatim: "at an FPR of 0.01%, URLTran_BERT has a TPR of 86.80% compared to 71.20% for URLNet and 52.15%

for Texception" (a 21.9% relative TPR improvement). This is the standard transformer baseline, and it implicitly pulls in **URLNet** (character+word CNN, Le et al. 2018) and **Texception** as second-tier baselines. ([Microsoft](#)) ([Towards Data Science](#))

- **URLBERT / urlBERT** — Li, Wang, Xu et al., arXiv 2402.11495 (2024), "Continuous Multi-Task Pre-training for Malicious URL Detection." Contrastive + adversarial pretrained URL encoder; AUC > 0.99 on phishing detection; its own baselines are RoBERTa, TextCNN, and classical ML (KNN, RF, CNN, LR, NB, GBM). ([arXiv](#)) ([GitHub](#))
- **DomURLs_BERT** — El Mahdaouy et al. (2024), pretrained BERT for malicious domains/URLs; baselines include CharCNN, CharGRU, CharLSTM and urlBERT, evaluated on ThreatFox + Kaggle Malicious URLs with 60/20/20 splits. ([ResearchGate](#)) ([ResearchGate](#))
- **PhishGNN** — Bilot, Geis, Hammi, *SECRYPT* 2022, pp. 428-435 (DOI 10.5220/0011328600003283). GNN over the website hyperlink graph plus hand-designed features; reported "99.7% prediction accuracy" (GCN2/GCN3/ClusterGCN variants), evaluated with 10-fold cross-validation. Note: PhishGNN requires hyperlink-graph features that are not always available, so some recent papers (e.g., KnowPhish) explicitly exclude it as inapplicable to their datasets — relevant if your data is URL/SMS-text only. ([ResearchGate + 3](#))
- **Reference-based visual/LLM detectors** — **Phishpedia** (USENIX Security 2021), **PhishIntention** (USENIX Security 2022), **DynaPhish** and **PhishLLM** (USENIX Security 2024), and **KnowPhish / KnowPhish Detector (KPD)** — Yuexin Li, Chengyu Huang, Shumin Deng, Mei Lin Lock, Tri Cao, Nay Oo, Hoon Wei Lim, Bryan Hooi, *USENIX Security* 2024, pp. 793-810. KnowPhish couples LLMs with multimodal knowledge graphs and was validated in a Singapore field study against state-of-the-art reference-based detectors. These matter if you process webpage content/screenshots; for a URL/SMS-text system they are context, not mandatory baselines. ([ACM Digital Library](#)) ([USENIX](#))

2. SMS / smishing detection baselines for low-resource / non-English (especially Bangla)

- **BanglaBERT** — Bhattacharjee, Hasan, Ahmad, Samin, Islam, Iqbal, Rahman, Shahriyar (BUET), *Findings of NAACL 2022*, pp. 1318-1327. Pretrained on "27.5 GB of Bangla pretraining data (dubbed 'Bangla2B+') by crawling 110 popular Bangla sites," establishing the first Bangla Language Understanding Benchmark (BLUB). This is the mandatory Bangla baseline. ([ACL Anthology](#))
- **XLM-RoBERTa** and **mBERT** — the multilingual transformer baselines. A 2024-2025 comparative Bengali SMS study found XLM-RoBERTa best at 98.92% accuracy, with BERT at 98.20% and RoBERTa at 94.41%; another study found BanglaBERT outperforming XLM-RoBERTa (~97% accuracy, 0.99 precision) on Bangla SMS. ([ResearchGate](#))
- **Hybrid BERT + character-level CNN** — Tanbhir, Afjal, Rahman, "Hybrid Machine Learning Model for Detecting Bangla Smishing Text Using BERT and Character-Level CNN," arXiv 2502.01518 (2025). 98.47% accuracy, attention layer, multiclass (smishing/promotional/normal). ([arxiv](#))

- **BanglaBERT + XGBoost** — hybrid on the BanglaBarta (Mendeley) Bangla SMS dataset; reports 99.08% accuracy, outperforming TF-IDF baselines. Also relevant: ELMo-based and RNN-based Bengali spam SMS detectors. ([ResearchGate](#))
- **English/U.S. line you must engage** — **SmishX** (SOUPS 2025; see §7) and **Nahapetyan, Prasad, Childs, Oest, Ladwig, Kapravelos, Reaves, "On SMS Phishing Tactics and Infrastructure," *IEEE S&P 2024*** (DOI 10.1109/SP54263.2024.00169), which mined >200M SMS from 11 public gateways, isolating 67,991 phishing messages into 35,128 campaigns and 600+ operations — the key data-source/threat-model citation. ([IEEE Xplore](#))

3. Android network malware / anomaly detection baselines

- **Drebin** — Arp, Spreitzenbarth, Hübner, Gascon, Rieck, *NDSS 2014*. Lightweight on-device static detection; the universal baseline.
- **MaMaDroid** — Onwuzurike, Mariconti, Andriotis, De Cristofaro, Ross, Stringhini, *NDSS 2017 / ACM TOPS 2019*. Markov chains over abstracted API-call sequences (family/package mode); F1 up to 0.99 in-period and robust over time; baseline comparison vs DroidAPIMiner. ([arxiv](#)) ([ACM Digital Library](#))
- **DroidEvolver** — Xu, Li, Deng, Chen, Xu, *EuroS&P 2019*. Self-evolving online-learning detector that updates without retraining or true labels using pseudo-labels; evaluated over ~68k apps across 6 years. ([Semantic Scholar](#))
- **Chen, Ding & Wagner, "Continuous Learning for Android Malware Detection," *USENIX Security 2023*** (arXiv 2302.04332). Hierarchical contrastive + active learning against concept drift; reduced false-negative rate from 14% (best baseline) to 9% and FPR from 0.86% to 0.48% across a 7-year span. This is the SOTA continual-learning baseline ("Chen-AL"), achieving ~92.4% F1 on the 7-year APIGraph dataset. ([arxiv](#))
- **Network-flow / anomaly specific** — deep learning on **CICAndMal2017** (DNN, 1D-CNN, BiLSTM, with LIME/SHAP explainability — *Journal of Information Security and Applications*, 2025); for general network anomaly/intrusion baselines: **NSL-KDD**, **UNSW-NB15**, **CICIDS2018** with CNN-LSTM, LSTM-AE, Bi-TCN/Bi-GRU baselines. Also cite **KronoDroid**, **DroidSieve** (obfuscation), and the **AndroZoo** repository as the data source.

4. Zero-day / drift evaluation protocol (what reviewers expect)

- **TESSERACT** — Pendlebury, Pierazzi, Jordaney, Kinder, Cavallaro, *USENIX Security 2019*. The mandatory protocol: enforce **C1 (temporal training-before-testing)**, **C2 (consistent goodware/malware ratio over time)**, **C3 (realistic ~10% malware spatial ratio in testing)**, and report **AUT (Area Under Time)** rather than a single inflated F1. Family-level holdouts and strict temporal splits are expected; mixing malware-family variants across train/test is a known fatal flaw. ([USENIX](#)) ([Ucl](#))
- **Newer expectations** — **Transcending TRANSCEND** (Barbero, Pendlebury, Pierazzi, Cavallaro, *IEEE S&P 2022*) for drift-aware classification; **Chow, D'Onghia, Linhardt, Kan,**

Arp, Cavallaro, Pierazzi, "Breaking Out from the TESSERACT" (2025) shows even TESSERACT-compliant evaluations on different datasets (APIGraph vs Transcendent) diverge, identifying five additional spatio-temporal bias factors; and LAMDA (longitudinal 12-year Android benchmark, arXiv 2505.18551, 2025), where Chen-AL's F1 drops from 92.4% (7-year) to 43% (12-year). For credible zero-day claims, evaluate on a longitudinal benchmark and report AUT. (ResearchGate)

5. Publishable novelty at Q1 — "genuinely novel" vs. "engineering combination"

The recurring reviewer distinction is between (a) a **new mechanism**, (b) a **new realistic evaluation regime**, or (c) a **genuinely under-served population/language/threat** — versus stacking known components. Evidence from the 2024–2026 multimodal-fusion literature:

- Pure feature-concatenation fusion is treated as incremental. Papers that get accepted claim a *specific* fusion innovation: **MIPDroid** ("modal independence preservation" with exclusive per-modality encoders + TCP-confidence dynamic weighting, *Information Fusion* 2026); **DMLDroid** (dynamic weighted fusion explicitly tested for resilience to *both* obfuscation and adversarial perturbation — noting "no study has systematically evaluated resilience against both," 97.98% accuracy, >98% under attack on CICMalDroid 2020); multimodal CNN+GCN ensembles that quantify the *gain over unimodal* (e.g., 83.1% + 80.6% → 88.3%). (ScienceDirect) (arxiv)
- Translation for your paper: **fusion of URL + SMS-text + on-device/network signals is not itself novel**. Novelty must come from (i) a new fusion/decision mechanism, (ii) robustness under temporal drift / obfuscation / adversarial SMS, or (iii) the Bangla low-resource + low-literacy user population with a real study and externally-grounded (not purely linguistic) reasoning. KnowPhish-style external grounding (brand/URL/webpage context) is a stronger novelty hook than another transformer fine-tune.

6. Minimum acceptable HCI / SOUPS / CHI user-study design

- **Qualitative / interview studies**: ~20–30 participants is acceptable. *Tabassum, Faklaris, Lipford, SOUPS 2024* used **N=29 semi-structured interviews** (the first SMiShing-susceptibility interview study). (USENIX)
- **Quantitative between-subjects warning experiments**: low hundreds, multiple conditions, random assignment. *Better Together* (CHI 2024) used **N=409**; a CHI 2023 phishing-warning study used **N=300**; an NDSS USEC 2025 vignette study used a 16-cell between-subjects design. Field experiments range into thousands (e.g., the Dutch Ministry study, >10,000 subjects). (ACM Digital Library + 3)
- **Usable AI-detection-system studies** (closest to your design): SmishX recruited **N=175 total** (a within-subject Study A and a separate AI-error Study B), reporting **SUS usability, detection accuracy before/after AI, trust, self-confidence, human-AI disagreement rate**,

and an inserted AI-error (false-negative) condition, with deliberate oversampling of older adults (40%). [\(illinois\)](#)

- **Expected metrics:** detection efficacy (accuracy / click-through), SUS (≥ 68 = above average; $\sim 80+$ = "excellent"), trust, comprehension, and subgroup analysis (age, literacy, technical experience). For a low-literacy Bangladesh population, add comprehension of icons/warnings and consider intermediated/shared-phone use (well-documented in South-Asia HCI: Sambasivan et al.; Ahmed et al. on shared phone use in Bangladesh). [\(illinois\)](#)

7. Most-cited recent ML + human-centered warning papers (2023–2025), especially non-English / developing-country

- **SmishX** — Yizhu Wang, Haoyu Zhai, Chenkai Wang, Qingying Hao, Nick A. Cohen, Roopa Foulger, Jonathan A. Handler, Gang Wang, "Can You Walk Me Through It? Explainable SMS Phishing Detection using LLM-based Agents," *SOUPS 2025*, pp. 37–56. This is your single most important comparison; details in §Details.
- **KnowPhish** — Li et al., *USENIX Security 2024* — LLM + multimodal knowledge graph, deployed and field-studied in Singapore. The leading "LLM + external knowledge" detection paper.
- **Tabassum, Faklaris, Lipford, SOUPS 2024** — "What Drives SMiShing Susceptibility?" (N=29). The human-factors anchor: users attend to content/format/links, not sender short-codes — a direct design input for warnings. [\(USENIX\)](#)
- **Better Together** — Berens, Schaub, Mossano & Volkamer, *CHI 2024* (paper 826) — between-subjects (n=409); the NoPhish video raised detection from 42.6%→86.9% and TORPEDO link-centric warnings from 43.31%→85.17%; tests redundancy between awareness videos and in-situ link warnings. [\(ACM Digital Library\)](#)
- **KorSmishing explainer** — Lee & Han, *EMNLP 2024* — a Korean-language LLM smishing system whose detection/explanation rely *solely on linguistic features* within SMS content. This is the closest non-English analog to your proposed system and the cleanest comparison point: a Bangla system that adds external grounding (URL/brand/webpage context) beyond linguistics is a clear contribution over both KorSmishing and SmishX.

Details: SmishX (your benchmark) in full

Detection performance and named baselines. SmishX reports overall accuracy **98.8%** (Phishing 100.0%, Spam 99.1%, Legitimate 98.2%). It was compared against two published systems whose code/models are available: **SpaLLM-Guard** (LLM-based; Salman et al., arXiv 2501.04985) at **96.8%** overall, and **PhishE** (ML-based; El Karhani et al., IEEE CSR 2023) at **85.0%** overall. Ablation baselines: "Baseline 1" (LLM directly on SMS without external context, mimicking existing LLM email-phishing methods) = 96.0%; "Baseline 2" (removing the spam prompt) = 89.7%. LLM backbone comparison: GPT-4o 98.8% > DeepSeek-v3 96.2% > Qwen2.5-

vl-72b 87.4%. Critically, neither PhishE nor SpaLLM-Guard produces explanations — explanation generation is SmishX's stated core contribution. (illinois + 4)

User study. N=175 total from Prolific (all U.S.-based): a within-subject **Study A (N=125)** and an AI-error **Study B (N=50)**. Study A participants judged 10 messages (5 phishing + 5 legitimate) before and after AI assistance. Detection accuracy improved **0.712 → 0.928** overall ($t=-15.01$, $p<0.001$); for **older adults (65+, 40% of sample)** **0.710 → 0.942**. Usability: **mean SUS 82.6** ("excellent"; older adults 85.3). Trust in AI rose $3.496 \rightarrow 4.264$; self-confidence $3.376 \rightarrow 3.640$. Human-AI disagreement occurred in 7.2% of assessments (more on legitimate SMS, 11.5%, than phishing, 2.9%), and the AI was correct in all disagreement cases. In Study B's inserted false-negative test, 48% initially flagged the phishing message correctly, but 18 of 24 reversed to "legitimate" after reading the (wrong) AI explanation — a documented over-reliance risk.

(illinois + 3)

Datasets. A re-labeled real-world set of **1,200 SMS** (622 legitimate + 578 malicious [259 phishing + 319 spam]; 53% contain URLs). Malicious messages sampled from Almeida et al. (SMS Spam Collection, DocEng 2011), Mishra & Soni (SoCPaR 2022), Nahapetyan et al. (IEEE S&P 2024), and Timko & Rahman (WISEC 2023). Released at github.com/yizhu-joy/SmishX. The authors acknowledge overfitting risk on this small dataset and ~60s/SMS latency as prototype limitations.

Language gap = your opportunity. SmishX is entirely English/U.S.-based; it does not test non-English handling and only contrasts itself with the Korean KorSmishing explainer. Its argument is that external context (URLs, brands, webpages) generalizes better than language-specific linguistic features — but this is **not empirically tested in another language**. A Bangla evaluation directly fills this hole.

Recommendations

Stage 1 — Detection baselines (mandatory in any submission):

- URL/phishing: URLTran, urlBERT, DomURLs_BERT (and, if you have graph/HTML features, PhishGNN; if webpage content, KnowPhish).
- Bangla/low-resource SMS: BanglaBERT and XLM-RoBERTa as primary; mBERT, the BERT+CharCNN hybrid (Tanbhir et al.), and a TF-IDF + classical ML floor.
- Android (if your fusion includes on-device/network signals): Drebin, MaMaDroid, DroidEvolver, and Chen et al. (USENIX Sec 2023) as the continual-learning SOTA.
- Report on at least one public Bangla SMS dataset (e.g., BangalaBarta/Mendeley) plus, if possible, a freshly collected Bangladesh dataset.

Stage 2 — Evaluation protocol: Adopt TESSERACT constraints (C1/C2/C3), report **AUT** alongside F1, and use **temporal splits + family/campaign-level holdouts**. For zero-day/drift

claims, evaluate longitudinally (cite LAMDA/Transcendent expectations). Add an **adversarial / obfuscation robustness** test, since that is now a differentiator.

Stage 3 — Positioning and novelty: Frame against the explainable-LLM-smishing line (**SmishX, KorSmishing explainer**) and the external-grounding line (**KnowPhish**). Claim novelty on (i) Bangla/code-mixed low-resource handling, (ii) external context grounding beyond linguistics, and (iii) a low-literacy Bangladeshi user population — not on "fusion" alone. State a concrete fusion *mechanism* (e.g., confidence-weighted decision fusion) rather than concatenation.

Stage 4 — User study: Minimum **N≈30 qualitative** (interview/comprehension, Tabassum-style) OR **N≈300+ quantitative between-subjects** for warnings (Better Together-style). If you build a usable detection+warning tool, replicate SmishX's design: before/after detection accuracy, **SUS**, trust, and an inserted AI-error condition, with explicit **low-literacy and older-adult subgroup analysis** and attention to shared/intermediated phone use.

Benchmarks that would change these recommendations: If you cannot beat BanglaBERT/XLM-RoBERTa on raw detection, pivot the contribution to explanation quality + user-study outcomes (where SmishX shows accuracy gains of ~0.21 are publishable even without SOTA raw detection). If your fusion shows no gain over the best unimodal baseline, drop the fusion framing and reposition as a single-modality Bangla explainable detector. If a Bangla SmishX-equivalent appears before submission, shift novelty to the low-literacy population and intermediated-use threat model, which remain unaddressed.

Caveats

- Many high-accuracy Bangla phishing results (98–99%) come from preprints, IEEE Access, MDPI, or workshop venues rather than Q1 security conferences, and rest on small or balanced datasets. Cite them as baselines but report their numbers with explicit dataset-size and class-balance caveats; do not present them as established Q1 results.
- SmishX's 98.8% and the BanglaBERT-family numbers are author-reported on small datasets (1,200 messages for SmishX) with acknowledged overfitting risk; treat as targets, not ground truth.
- PhishGNN and KnowPhish require features (hyperlink graphs, webpage screenshots/brand knowledge) that a URL/SMS-text-only system cannot supply; if your inputs differ, state explicitly why these are not applicable rather than omitting them silently — this is standard practice (KnowPhish itself excluded PhishGNN for this reason).
- The "fusion" novelty bar is rising fast: 2026-dated fusion papers already demand dual obfuscation+adversarial robustness testing. Budget for those experiments or reframe the contribution around language/population.
- TESSERACT compliance is necessary but, per the 2025 "Breaking Out from the TESSERACT" work, no longer sufficient on its own; anticipate reviewer scrutiny on dataset

provenance (AndroZoo sampling, label stability) and consider longitudinal evaluation.